Dermatologic Diagnosis with Machine Learning

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Please note all the figures are attached to the Appendix. Link to github code -

https://github.com/rahul-purswani/skin-disease-classifier

Abstract

Skin diseases are prevalent globally, affecting millions each year. The similarity in appearance across a wide spectrum of these conditions poses significant challenges for accurate diagnosis. Early and precise detection is crucial for effective treatment and patient management. This project introduces an automated system that leverages the HAM10000 dataset, a comprehensive collection of dermatoscopic images, to enhance the diagnostic accuracy of skin diseases. By developing a deep learning model capable of distinguishing between various skin lesions with high accuracy. The system aims to mitigate the risk of misdiagnosis and optimize patient outcomes, highlighting the transformative potential of automated systems in dermatological practice. The findings underscore the benefits of machine learning in medical imaging and pave the way for future advancements in the field of dermatology.

Related Work

Skin lesion classification using deep learning has become a significant research domain due to the critical need for early and accurate diagnosis of skin cancer types such as melanoma, which is the most lethal form of skin cancer. Recent advancements have leveraged deep learning technologies to enhance diagnostic accuracy and efficiency.

One notable approach in the field is the use of Convolutional Neural Networks (CNNs) to analyze dermoscopic images for skin cancer detection. Studies like Alam Milton's work on an ensemble of deep neural networks, specifically for the ISIC 2018 challenge, have shown promising results in improving the classification of skin lesions by integrating multiple CNN architectures like PNASNet-5-Large, InceptionResNetV2, and SENet154 (ar5iv). Another study implemented an ensemble of hybrid models using ResNet-50 and DenseNet-201 architectures, which significantly improved the balanced accuracy of skin lesion classification (ar5iv).

Dataset and Pre-processing

In this project, we use the HAMS 10000 dataset (sourced directly from Harvard website). The dataset contains ~10000 dermatoscopic images from the University of Queensland and the Vienna General Hospital. It features a variety of lesions including melanoma, basal cell carcinoma, and others, richly annotated with expert-reviewed diagnoses and metadata such as lesion type, localization, patient age, and sex.

The input for our models is the images of the lesions and the output is lesion_type. In pre-processing, there are some missing values in the metadata, and we just delete those records because we have plenty of data. We also resize all the images (as per the model requirements), and perform random transformations (like vertical/horizontal flip, rotation, etc). A sample of these transformations in augmentation is shown in Figure 8. Lastly, all the lesion_types are encoded from strings to vectors (using one-hot encoding).

The training set originally contains 7511 records and the testing set contains 2504 records (75% training and 25% testing, stratified). However, there is a heavy class imbalance in lesions, as shown in Figure 3. To address this problem, we do some random undersampling for nv and some over sampling for the rest of the lesion types on the training set. This way of augmentation works because we are also applying random transformations to the images before feeding them to the model as shown in Figure 8. After augmentation, the training set contains 26149 records and the testing set contains 2504 records.

Disease Distribution

The core of the dataset analysis revolves around the distribution of different skin lesions. Using bar charts, each lesion type's frequency is displayed as shown in Figure 3, showing a diverse range of conditions from common benign lesions to rare malignant cases. This diversity is key to training a model that can accurately identify and differentiate between multiple types of skin conditions. In the dataset, the highest or common disease type is ‘nv’ which is Melanocytic nevi and Dermatofibroma has the lowest frequency.

Age Distribution

The age distribution of subjects in the dataset is analyzed to understand the demographic spread. In Figure 1, we can see a wide variation in age from infancy to adulthood with highest frequency for the age of 40-60.

Sex Distribution

The dataset is fairly balanced in terms of distribution of male and female subjects as shown in Figure 2. This balance helps in mitigating bias in the model training process, ensuring that the classification performance is not skewed towards one gender. The data has 54% of male in comparison to 45.5% female alongside some minor proportions of unknown groups.

Localization Distribution

Localization of lesions, or the body areas affected, is another focus of the dataset analysis. Certain skin conditions are more likely to appear on specific parts of the body as shown in Figure 4, and understanding this distribution aids in enhancing the model’s diagnostic accuracy. Top 5 of the common local body areas are the back, lower extremity, trunk, upper extremity and the abdomen. Moreover, as shown in Figure 5, males have the highest occurrence of lesions in the back while females have lesions in their lower extremity body region.

Augmentation Analysis

We applied random flip augmentations to our images and also applied image color enhancement to give our model exposure to diversified data. Figure 8 shows random flips from -25 to 25 degree and color improvement using the PIL library. This shows a sample of augmentations on images, before being fed into the model.

Methodology

Custom CNN Model

In our first approach (our baseline), we built a custom CNN model to classify the skin lesions. We developed two different versions of CNN (can be seen on training.ipynb), both containing combinations of convolution, pooling, and linear layers. We then picked out the best performing model (as per the accuracy on the training and testing sets). Figure 13 shows the model architecture for the best performing CNN model. First the image passes through multiple convolution layers to extract features from the input image and we also apply some pooling, followed by linear layers to classify based on the features extracted.

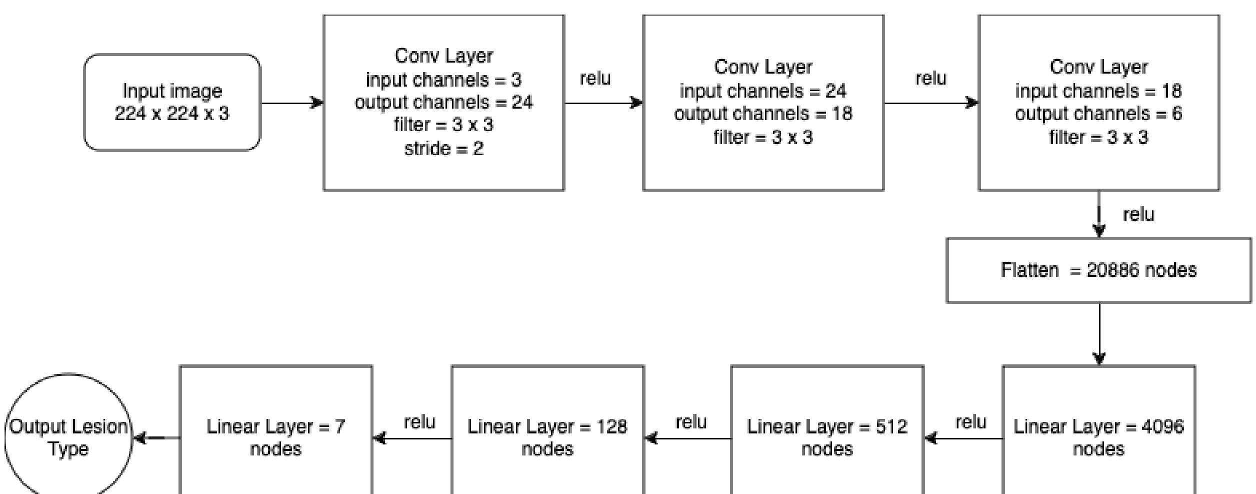

Figure 13 : Custom CNN model architecture

Major strength of this architecture is that it allows for feature extraction and that significantly improves the classifications. However, the training process can be computationally expensive, and requires a very large amount of data.

Transfer Learning

In our second approach, we used transfer learning to fine-tune state of the art models, pre-trained models, to classify different skin lesions. We fine-tuned ResNet18, ResNet50, Efficientnet_b1, MobileNet_V3_Small, and ViT_base_16_224. ResNet18, ResNet50, Efficientnet_b1, and MobileNet_V3_Small are known to work best for image classification tasks and they all contain convolution layers, bypass layers, pooling layers, and fully connected layers with variations in configuration and activation functions. ViT_base_16_224 contains patch embeddings, positional embeddings, and a transformer encoder, followed by a classification token. We added a final classification layer to each of the models with 7 nodes, to fit the HAMS dataset.

Major strength of this approach is that it allows us to use a much more robust model that performs significantly better than the custom CNN models. However, since every model has a different configuration for data and labels as input, the training process become complicated and is computationally very expensive

Experiments

Training Configuration

- The dataset was split into 75% for training and 25% for testing, using stratification to preserve class balance and reduce bias. We then did oversampling on the training set (as explained above) and finally had 26149 images for training and 2504 for testing.
- In pre-processing, all the labels were converted into vectors using one-hot encoding. All the images were resized to 224 x 224, as per the model training requirements. Random augmentations (such as random flips, rotations, etc.) were applied to training images to make the model generalize better and reduce the risk of overfitting.

- Training was executed on P100 GPUs, leveraging that CUDA technology to enhance computational speed and efficiency.
- Each model was trained for 10 epochs, due to limited hours on the GPU. It took an average of ~2.5 hours for one model to complete 10 epochs of training. The learning rate was set to 0.0001 and 0.0005 for the ViT model. We experimented with some more learning rates but decided to stick with the standard. The batch size was set to 128 for all the models and 64 for the ViT model.
- For all the models, we chose Adam as the optimizer and cross entropy loss for the loss function. We also experimented with Stochastic Gradient Descent as the optimizer, but found Adam to give better results in terms of accuracy.
- Post training, GPU cache was cleared for the next model to be trained.

Training Loop

To initiate the model's training phase, we set the model to training mode. Utilizing a custom dataset in PyTorch, we retrieve preprocessed labels and augmented images. It is important to note that random augmentations are only applied to the training set, images from the testing set are just resized for the model. Next, we perform a forward pass on the batch and compute the loss. This loss is then propagated backward to update the model parameters. Concurrently, we track and update the running accuracy and loss statistics for the current epoch. Following this, the model is switched to evaluation mode to generate predictions on the test set, and we update the testing accuracy for the epoch. Predictions on the test set help us evaluate if the model is overfitting or underfitting the training data. This process is repeated for every epoch and for each model. There were some minor tweaks made particular to the model. Lastly, graphs were plotted for training accuracy, testing accuracy, and training loss.

Model Evaluation

Model was mainly evaluated in terms of accuracy on the training and testing sets as explained above. Table 1 shows the training, testing accuracy and training loss. Accuracy and loss graphs are shown in Figure 6-7 and Figure 9 - 12. We chose our custom CNN as baseline, because it represents a simpler architecture of all the other models we have used. We found the EfficientNet_B1 model to perform the best with training accuracy of 98.54%, testing accuracy of 89.70%, and training loss of 0.042. Almost all the models are somewhat overfitting the training data due to the class imbalance (as explained above, Figure 3), and to mitigate this we are already oversampling and augmenting on training data (as explained in above sections). Without these measures, the model performances were far worse. However, we can include more measures like early stopping, to improve the model generalization. As seen in evaluation graphs for models, we can stop training at epoch 6-7 for some of the models.

Model Architecture	Training Accuracy	Testing Accuracy	Training Loss
Custom_CNN	94.20%	63.50%	0.165
ResNet18	94.91%	80.83%	0.140
ResNet50	93.73%	80%	0.175
Efficientnet_b1	98.54%	89.70%	0.042
Mobilenet_v3_small	97.91%	81.35%	0.061
ViT_base_patch16_244	83.16%	63.78%	0.432

Table 1: Evaluation results for each model.

Conclusion

This study effectively demonstrated the utilization of advanced deep learning methodologies, particularly Convolutional Neural Networks (CNNs) and transfer learning, for classifying skin lesions with the HAM10000 dataset. By employing various custom CNN architectures, such as ResNet18, ResNet50, EfficientNet_b1, MobileNet_V3_Small, and ViT_base_16_224, the project achieved significant outcomes, underscoring the potential and effectiveness of these models in medical imaging tasks.

Key Findings:

- The EfficientNet model outperformed others by achieving the highest training and testing accuracies, approximately 98% and 87% respectively, along with the lowest loss. However, it notably underperformed in identifying Dermatofibroma and vascular skin lesions, which underscores some limitations in feature generalization for specific conditions.

- More complex models such as ResNet50 and the Vision Transformer exhibited overfitting, evident from their very high training accuracy and very low validation accuracy. This suggests a need for further tuning to balance model complexity with generalization across unseen data.

Implications for Future Work:

We successfully built a model that performs well in classifying skin lesions, showcasing the effectiveness of advanced deep learning techniques in this medical imaging task. Future studies could explore:

- Combining predictions from multiple models to improve accuracy and robustness. We think a multimodal approach training on parameters specific to lesion (like existing medical conditions), combined with the output from the image can improve the model accuracy by a lot.
- We would also like to research further on smarter ways of augmenting data to improve the model performance in classifying skin lesions. We did some texture and edge analysis on the lesions, but would like to incorporate data preprocessing, and investigate the effect on model performance.
- Enhanced Regularization and Dropout Techniques: Investigating more advanced regularization strategies and dropout techniques to better manage overfitting, which creates the gap between training and validation accuracies.

References

1. Tschandl, P., Rosendahl, C., Akay, B. N. (2019). "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions." Scientific Data, 6, Article number: 194.
2. Esteva, A., Kuprel, B., Novoa, R. A., et al. (2017). "Dermatologist-level classification of skin cancer with deep neural networks." Nature, 542(7639), 115–118.
3. Milton, Md Ashraful Alam. "Automated skin lesion classification using ensemble of deep neural networks in isic 2018: Skin lesion analysis towards melanoma detection challenge." arXiv preprint arXiv:1901.10802 (2019).
4. Ali, Redha, and Hussin K. Ragb. "Skin lesion segmentation and classification using deep learning and handcrafted features." arXiv preprint arXiv:2112.10307 (2021).

Appendix

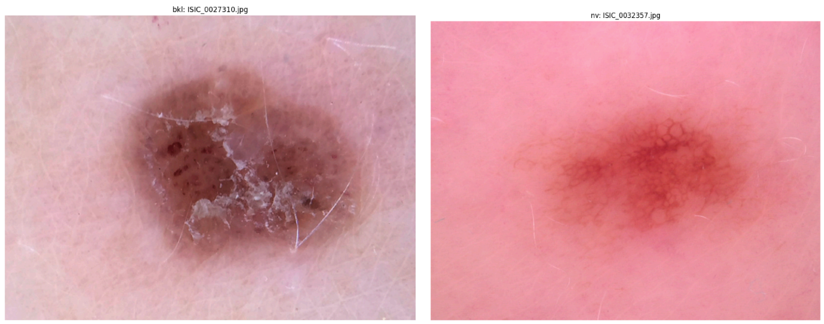

Figure 0: Two Skin Lesions

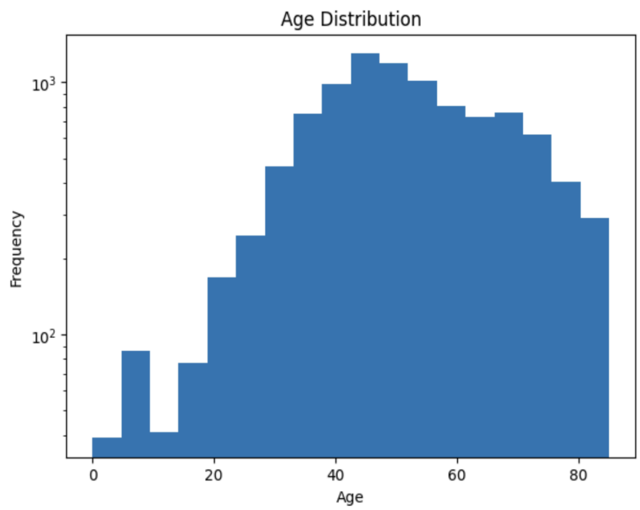

Figure 1: Age Distribution

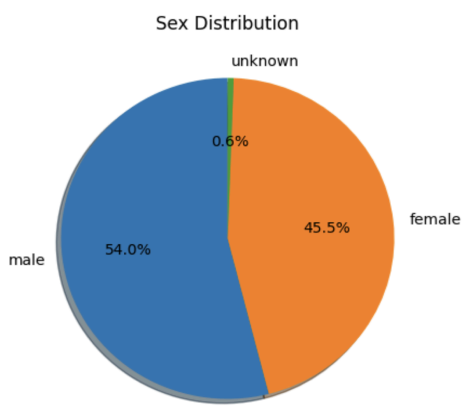

Figure 2: Sex Distribution

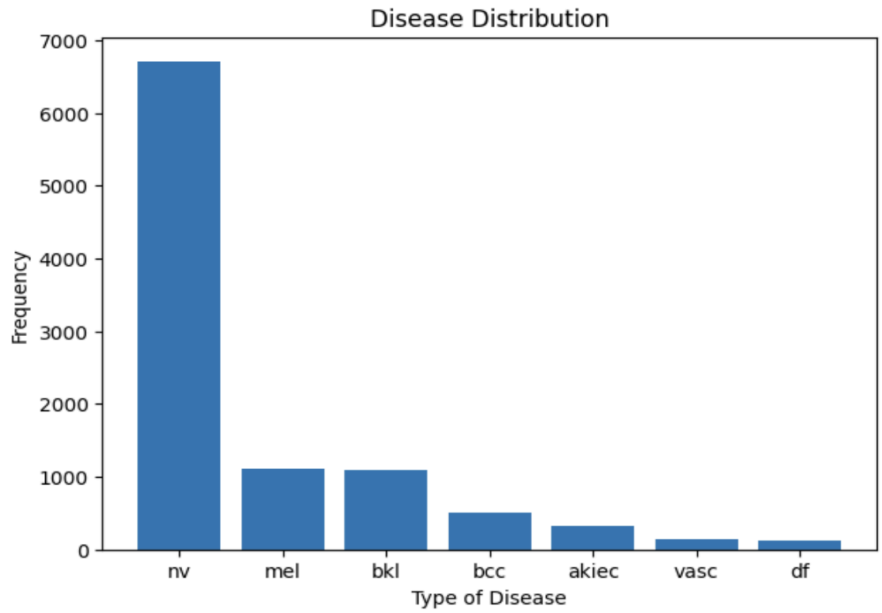

Figure 3: Disease Distribution

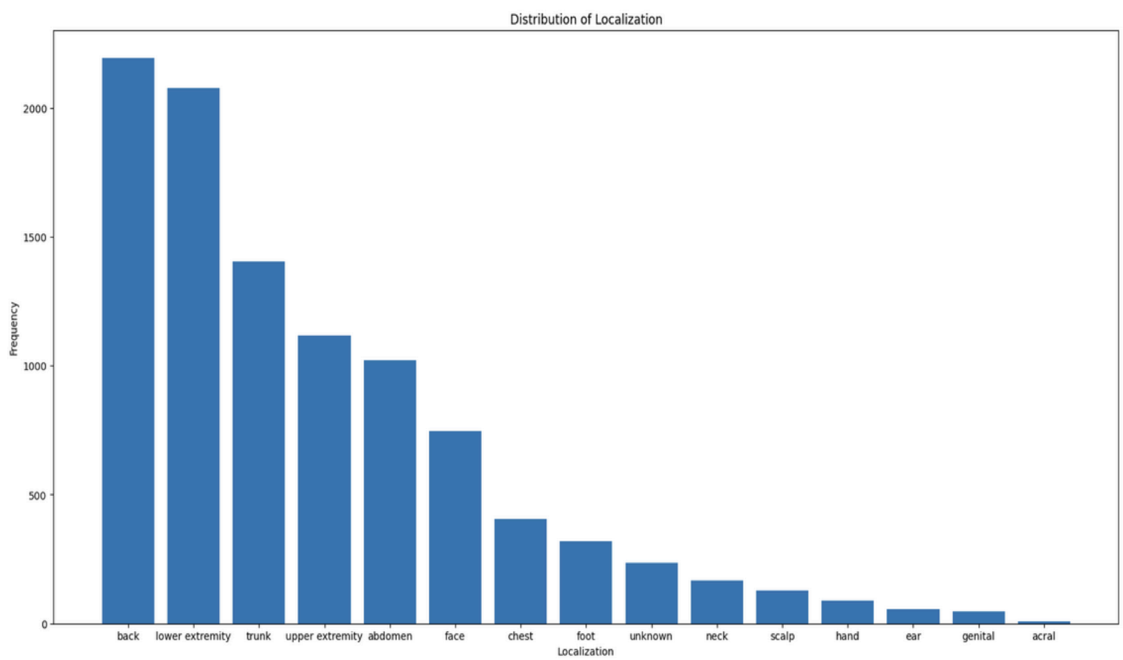

Figure 4: Localization Distribution

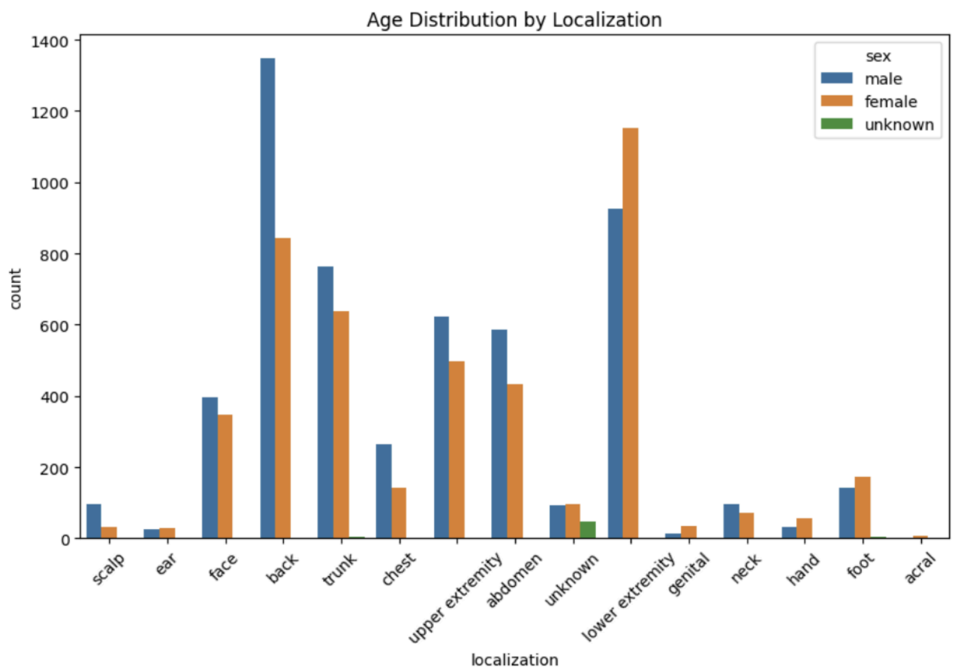

Figure 5: Sex Distribution with Localization

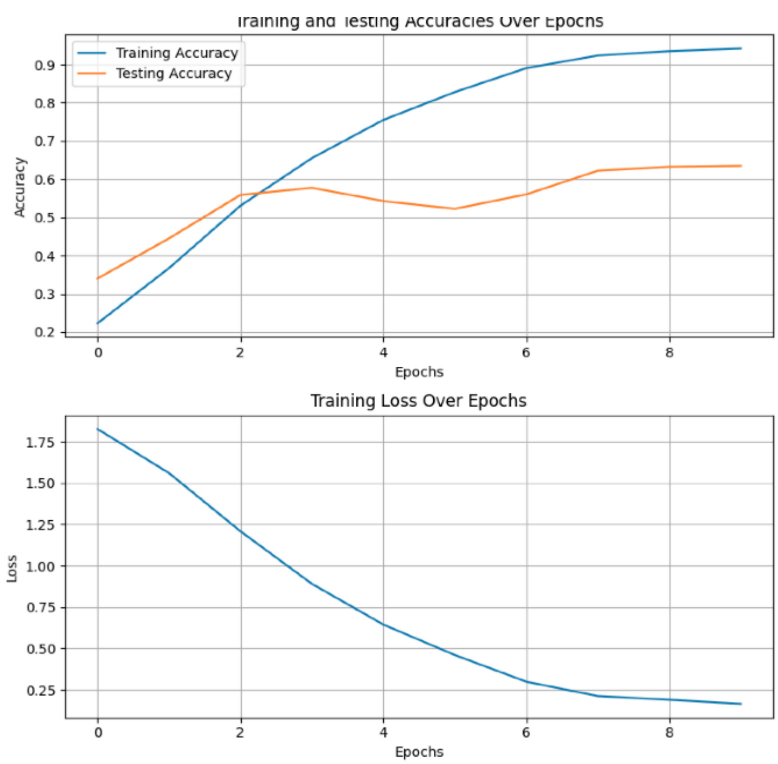

Figure 6: Custom Architecture Accuracy

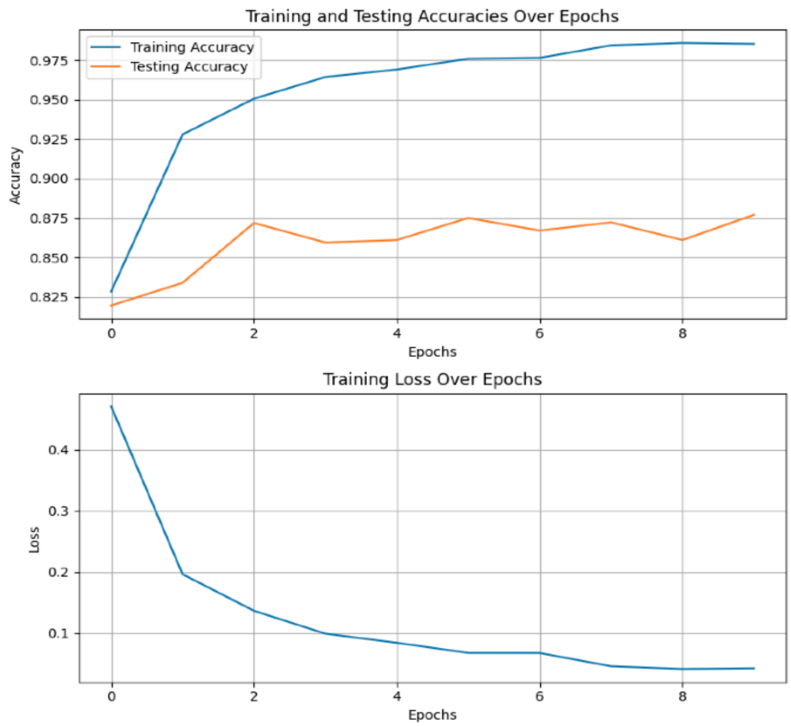

Figure 7: Efficient Net Accuracy

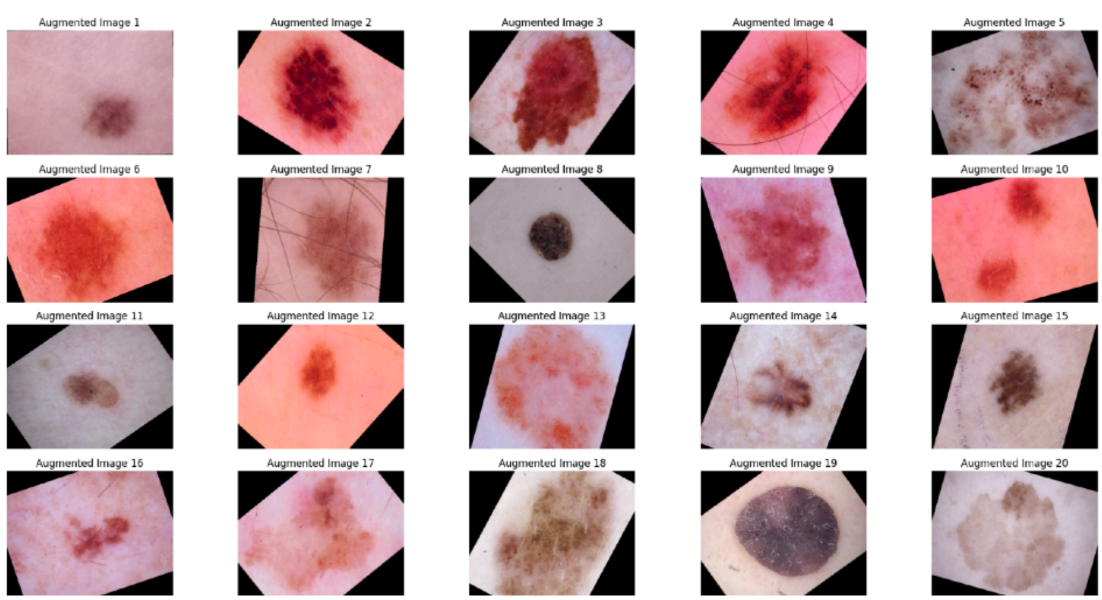

Figure 8: Augmentation Analysis

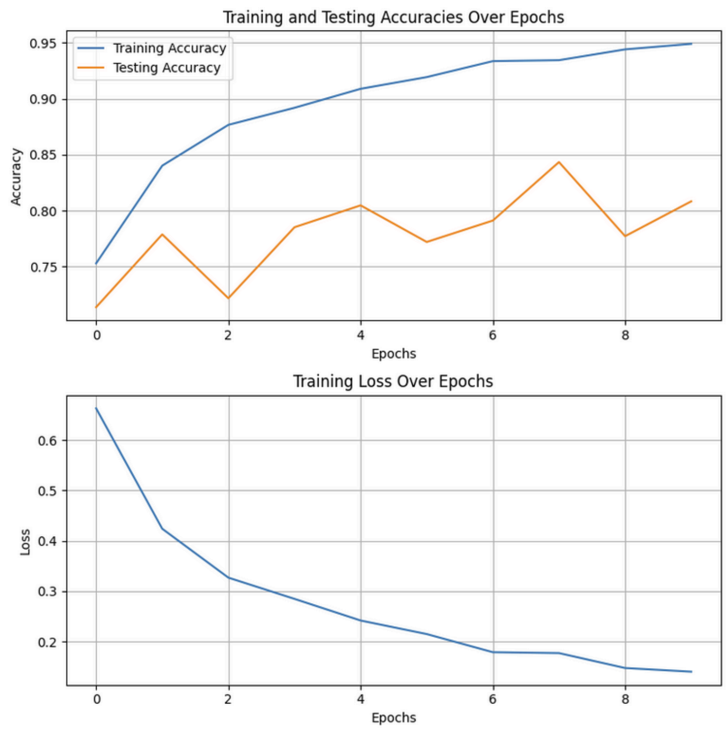

Figure 9: Accuracy and loss graphs for Resnet18.

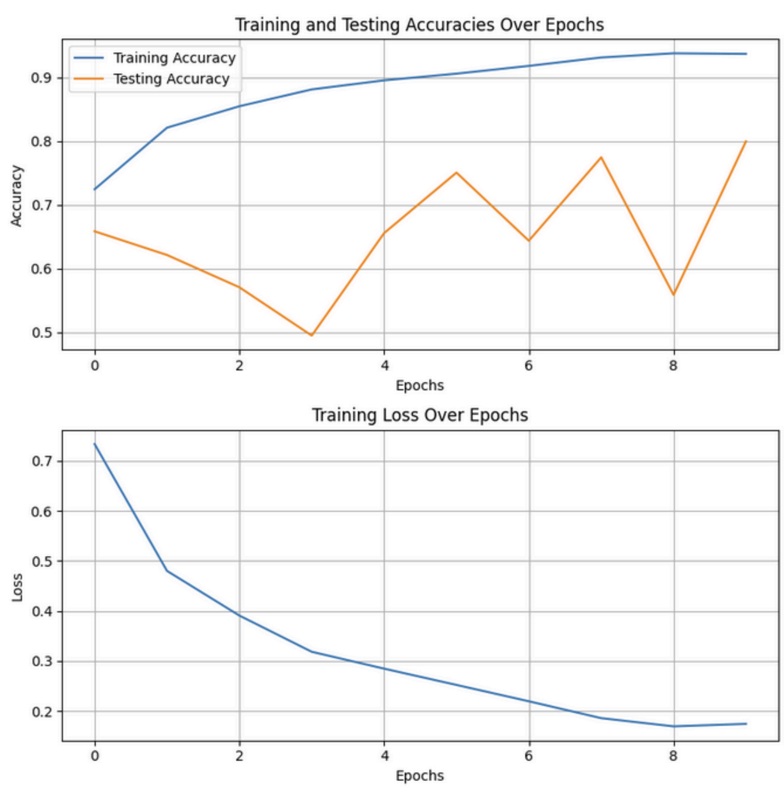

Figure 10: Accuracy and loss graphs for Resnet50

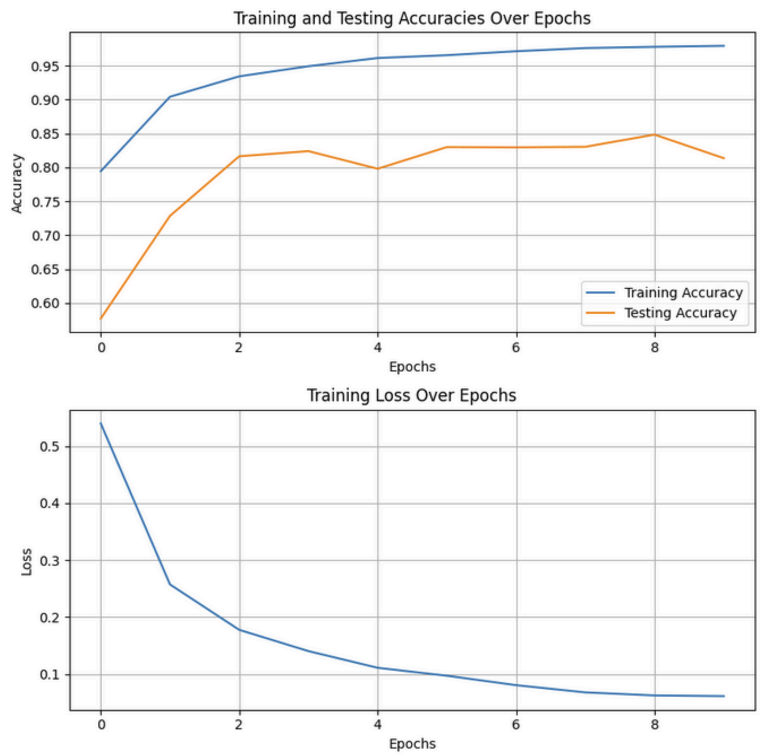

Figure 11: Accuracy and loss graphs for Mobilenet (v3_small).

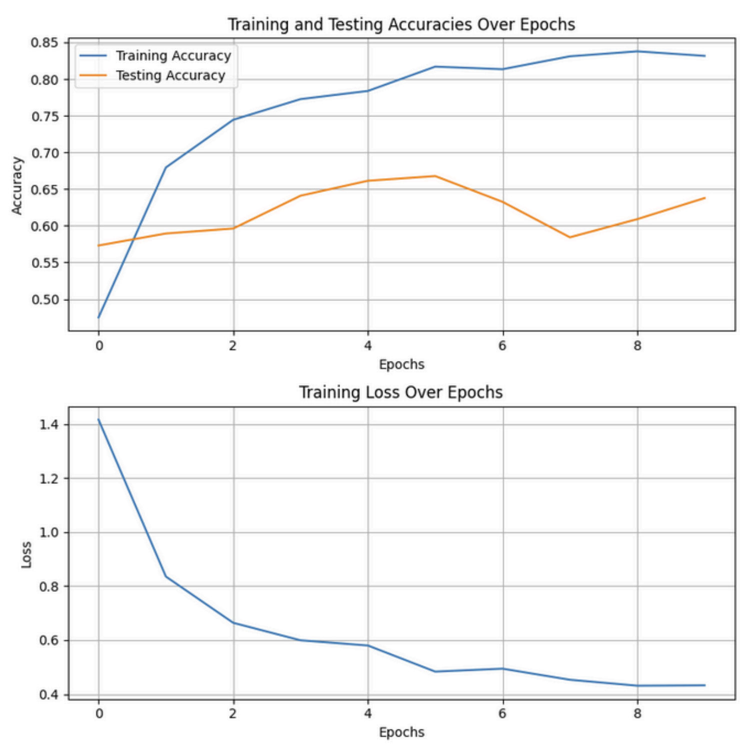

Figure 12: Accuracies and Loss graphs for Vision Transformer